

RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF TECHNOLOGY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 3219 – Mobile Application Development

Practical Sheet – 04

Widgets, Event Handling & the Android Project Structure

Objectives

- Understand the role of build.gradle, AndroidManifest.xml and activity_main.xml in an Android project
- Connect Java code to XML widgets using findViewById()
- Handle button click events using View.OnClickListener
- Read user input, validate it, and perform a calculation in Java
- Build, run and test a simple app on an emulator or device

Task 1: Project Setup

Step 1: Create New Project

- Open Android Studio
- Select File → New → New Project
- Choose Empty Views Activity template
- Name: BMI Calculator
- Package name: com.example.bmicalculator
- Language: Java
- Minimum SDK: API 24 (Android 7.0)

Step 2: Configure build.gradle (Module: app)

This file tells Android Studio what your app is called, which SDK versions to target, and which libraries (dependencies) it needs.

```
plugins {
    alias(libs.plugins.android.application)
}

android {
    namespace = "com.example.bmicalculator"
    compileSdk {
        version = release(36) {
            minorApiLevel = 1
        }
    }

    defaultConfig {
        applicationId = "com.example.bmicalculator"
        minSdk = 24
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            optimization {
                enable = false
            }
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

dependencies {
    implementation(libs.activity.ktx)
    implementation(libs.appcompat)
    implementation("androidx.constraintlayout:constraintlayout:2.1.4")
    implementation(libs.constraintlayout)
    implementation(libs.material)
    testImplementation(libs.junit)
    androidTestImplementation(libs.espresso.core)
    androidTestImplementation(libs.ext.junit)
}
```

Click "Sync Now" when prompted at the top of the editor so Gradle downloads the dependencies.

Task 2: Configure AndroidManifest.xml

The manifest declares your app's components — here, just one Activity — and marks MainActivity as the launcher screen (the one that opens first).

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
```

```
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.BMICalculator">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:windowSoftInputMode="adjustResize">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Task 3: Design the UI (activity_main.xml)

Build the layout below in res/layout/activity_main.xml. It uses a vertical LinearLayout with two EditText fields for input, a Button to trigger the calculation, a TextView to show the result, and a Reset button.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="BMI Calculator"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="16dp"/>

    <EditText
        android:id="@+id/etWeight"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Weight (kg)"
        android:inputType="numberDecimal"/>

    <EditText
        android:id="@+id/etHeight"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Height (m)"
        android:inputType="numberDecimal"
        android:layout_marginTop="12dp"/>

    <Button
        android:id="@+id/btnCalculate"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Calculate BMI"
        android:layout_marginTop="20dp"/>
```

```
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text=""
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginTop="24dp"/>

    <Button
        android:id="@+id/btnReset"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Reset"
        android:layout_marginTop="12dp"/>

</LinearLayout>
```

Note the android:id attributes — these are how MainActivity.java will find and control each widget.

Task 4: Implement MainActivity.java

This class links to the XML widgets, listens for button clicks, and calculates the BMI (Body Mass Index) from the entered weight and height.

```
package com.example.bmicalculator;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    // UI Components
    private EditText etWeight, etHeight;
    private Button btnCalculate, btnReset;
    private TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Link variables to XML widgets using findViewById
        etWeight = findViewById(R.id.etWeight);
        etHeight = findViewById(R.id.etHeight);
        btnCalculate = findViewById(R.id.btnCalculate);
        btnReset = findViewById(R.id.btnReset);
        tvResult = findViewById(R.id.tvResult);

        // Event handling: Calculate button
        btnCalculate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                calculateBmi();
            }
```

```
        });

        // Event handling: Reset button (lambda style)
        btnReset.setOnClickListener(v -> resetFields());
    }

    private void calculateBmi() {
        String weightStr = etWeight.getText().toString().trim();
        String heightStr = etHeight.getText().toString().trim();

        if (weightStr.isEmpty() || heightStr.isEmpty()) {
            Toast.makeText(this, "Please enter both weight and height",
Toast.LENGTH_SHORT).show();
            return;
        }

        try {
            double weight = Double.parseDouble(weightStr);
            double height = Double.parseDouble(heightStr);

            if (height <= 0 || weight <= 0) {
                Toast.makeText(this, "Values must be greater than zero",
Toast.LENGTH_SHORT).show();
                return;
            }

            double bmi = weight / (height * height);
            String category = getBmiCategory(bmi);

            tvResult.setText(String.format("BMI: %.1f (%s)", bmi, category));

        } catch (NumberFormatException e) {
            Toast.makeText(this, "Please enter valid numbers",
Toast.LENGTH_SHORT).show();
        }
    }

    private String getBmiCategory(double bmi) {
        if (bmi < 18.5) return "Underweight";
        else if (bmi < 25) return "Normal";
        else if (bmi < 30) return "Overweight";
        else return "Obese";
    }

    private void resetFields() {
        etWeight.setText("");
        etHeight.setText("");
        tvResult.setText("");
    }
}
```

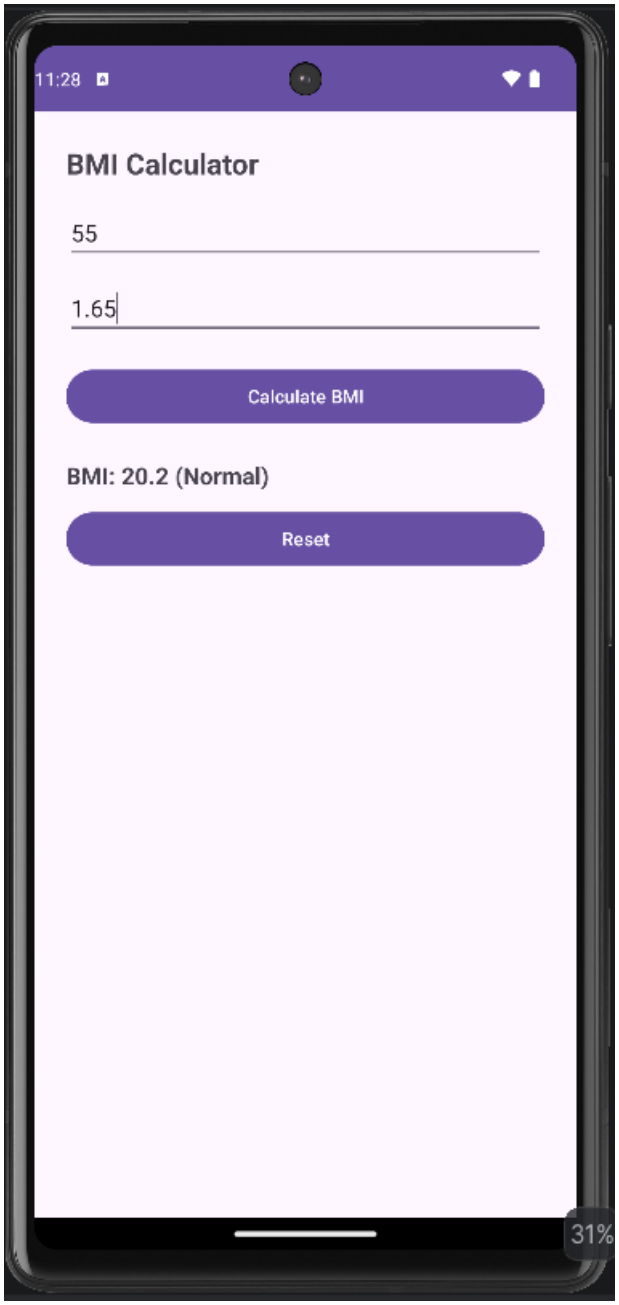
Walk through:

- findViewById() — connects a Java variable to an XML widget by its id
- setOnClickListener() — registers code to run when a button is tapped (this is event handling)
- try/catch with NumberFormatException — handles invalid or non-numeric input gracefully
- String.format() — formats the BMI result to one decimal place

Task 5: Run and Test the App

Step 1: Run the App

- Select an emulator or connected device from the dropdown
- Click Run (green triangle)



Step 2: Test the Following Cases

- Enter valid weight and height → verify correct BMI and category appear
- Leave a field empty and tap Calculate → verify the warning Toast appears
- Enter text instead of numbers → verify the "valid numbers" Toast appears
- Enter 0 for height → verify the app does not crash (division-by-zero guard)
- Tap Reset → verify all fields clear

Task 6: Debugging Common Issues

Issue 1: App crashes immediately on launch

- Check that every id used in findViewById() exactly matches an id in activity_main.xml

Issue 2: NumberFormatException even with numbers entered

- Check the EditText inputType is numberDecimal and that .trim() is applied before parsing

Issue 3: Nothing happens when a button is tapped

- Confirm setOnClickListener() is attached in onCreate(), after findViewById()